Understand, Control and Reduce Variation

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Hospira Costa Rica

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About Hospira Costa Rica



About Hospira Costa Rica

- > 2800 Employees
- Operated 24/7
- 3 Products:
  - Gravity Sets
  - Injectable Syringes
  - Infusion Pumps
- > 1000 Validated Items (Equipment/Processes/Systems, etc)
- **A lot of variation sources to be controlled and improved!**



"After living with their dysfunctional behavior for so many years (a sunk cost if ever there was one), people become invested in defending their dysfunctions rather than changing them."
— Marshall Goldsmith
Mojo

“A state of statistical control is not a natural state for a manufacturing process; it is instead an achievement, arrived at by eliminating one by one, by determined effort, the special causes of excessive variation."
— W. Edwards Deming

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About the presentation

- Presenter is no statistician or statistical process control expert.
- Presenter has worked primarily for MD manufacturing, not pharma or development. – May not be able to address very specific items on these.
- Theoretical approach – showing which tools may be used, but now how to use the tools.
- Why is a guy from MD manufacturing talking about new Process Validation guideline?
  - Hospira uses same validation approach for drugs and devices sites aligned with the new process validation guideline.
  - General Process Validation principles for drugs are similar to the ones from MD manufacturing.
- Could've talked about this forever! Summary of main elements and ideas.
- Questions or input anytime

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Definition and Sources of Variation identified in Validation Processes

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What is variation?

- Variation (or variability) is everywhere, all the time, in everything, the “natural state”.
- Variation is the deviation you have from an ideal or required state.
- Divided in two parts:
  - Common Cause (Natural, Chance Cause) Variation-> This is expected to be in a process. It requires a team approach towards identifying ways of improving the process.
  - Special Cause -> Unusually good or bad trend in a process. This is not expected in a process and requires investigation and correction.

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What is process control and capability?

- State of control: A condition in which the set of controls consistently provides assurance of continued process performance and product quality. (ICH Q10)
  - Process monitored is not subject to any large unusual influence.
  - The major characteristics of consecutive items produced by the process belong to the same statistical distribution, but are statistically independent from each other
- Capability of a process: Ability of a process to produce a product that will fulfill the requirements of that product. The concept of process capability can also be defined in statistical terms. (ISO 9000:2005)

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What is understanding a process?

- A process is generally considered well understood (according to FDA) when:
  - all critical sources of variability are identified and explained;
  - variability is managed by the process; and,
  - product quality attributes can be accurately and reliably predicted over the design space established for materials used, process parameters, manufacturing, environmental, and other conditions. The ability to predict reflects a high degree of process understanding.

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Possible sources of variation – the M's

- Instruments → Measurement
- Equipment → Machine
- Facilities and Utilities →Machine
- Processes → Man
- Procedures → Method
- Computer Systems / Software → Machine
- Product Design Changes → Material / Method
- Raw Material → Material
- Temperature, Humidity, Light → "Mother Nature" (Environment)

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Possible sources of variation – the M's

- Vibration→ Machine / "Mother Nature"
- Purity of process materials → Material
- Human Factors (training, ergonomic factors, stress, etc) →Man
- Wear of equipment → Machine

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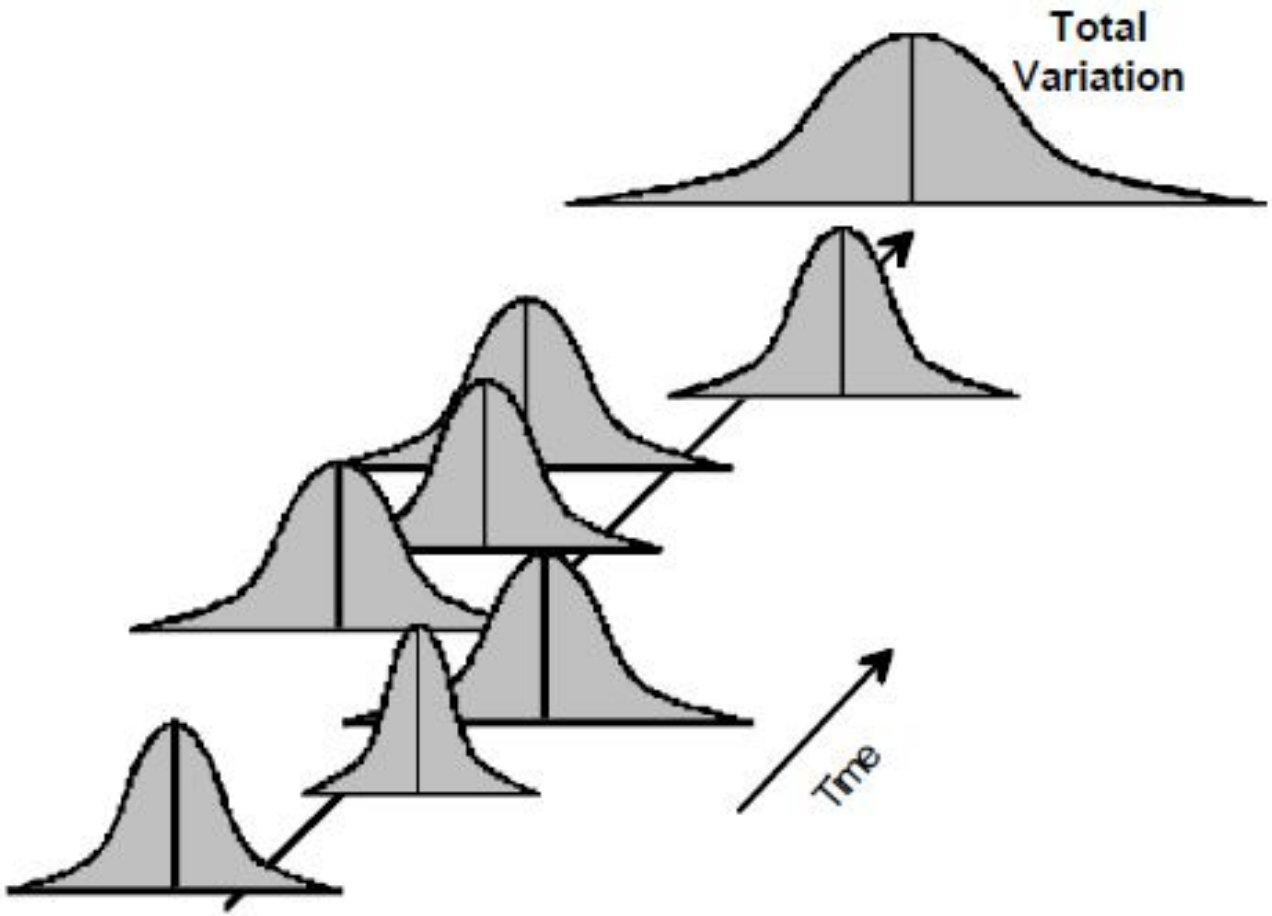
Components of Variation in a Process

- Total Process Variation
- Lot to Lot Variation
- Within Lot Variation
- Stream to Stream Variation
- Within Stream Variation
- Time to Time Variation
- Within Time Variation
- Part to Part Variation
- Within Part Variation
- Measurement Error
- Inherent Process Variation



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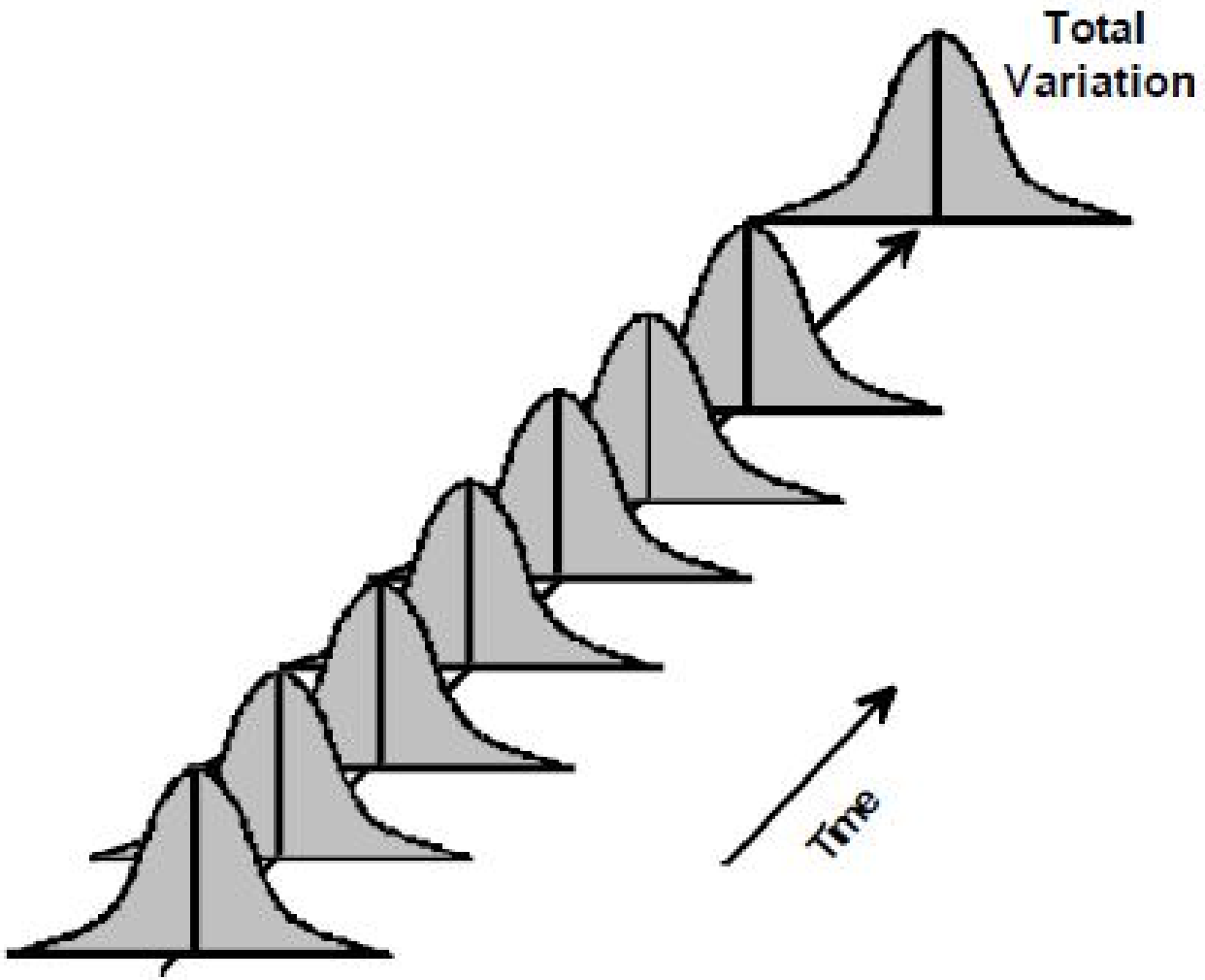
Unstable Process



Taken from GHTF/SG3/N99-10:2004 (Edition 2) - Quality Management Systems - Process Validation Guidance

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Stable Process



Taken from GHTF/SG3/N99-10:2004 (Edition 2) - Quality Management Systems - Process Validation Guidance

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Regulatory guidance related to variation control

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Regulatory Guidance for Medical Devices

- GHTF Medical Device PV Guidance
  - The product should be designed robustly enough to withstand variations in the manufacturing process and the manufacturing process should be capable and stable to assure continued safe products that perform adequately.
  - Process and product data should be analyzed to determine what the normal range of variation is for the process output. Knowing the normal variation of the output is crucial in determining whether a process is operating in a state of control and is capable of consistently producing the specified output.

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Regulatory Guidance for Medical Devices

- GHTF Medical Device PV Guidance
  - Appropriate measures should be taken to eliminate controllable causes of variation. Eliminating controllable causes of variation will reduce variation in the process output and result in a higher degree of assurance that the output will consistently meet specifications.
  - Nonconformities often occur because of errors made and because of excessive variation. Obtaining a process that consistently conforms to requirements requires a balanced approach using both mistake proofing and variation reduction tools.

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Regulatory Guidance for Pharma

- Part 210-211 – Drugs → Validation requirements
  - The CGMP regulations for validating pharmaceutical (drug) manufacturing require that drug products be produced with a high degree of assurance of meeting all the attributes they are intended to possess (21 CFR 211.100(a) and 211.110(a)).
  - The foundation for process validation is provided in § 211.100(a), which states that "[t]here shall be written procedures for production and process control designed to assure that the drug products have the identity, strength, quality, and purity they purport or are represented to possess..." (emphasis added). This regulation requires manufacturers to design a process, including operations and controls, which results in a product meeting these attributes.

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Regulatory Guidance for Pharma

- Part 210-211 – Drugs → Sampling and Monitoring
  - control procedures ". . . be established to monitor the output and to validate the performance of those manufacturing processes that may be responsible for causing variability in the characteristics of in-process material and the drug product"
  - CGMP regulations regarding sampling set forth a number of requirements for validation: samples must **represent the batch** under analysis (§ 211.160(b)(3)); the sampling plan must **result in statistical confidence** (§ 211.165(c) and (d)); and the batch must **meet its predetermined specifications** (§ 211.165(a)).

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Regulatory Guidance for Pharma

- Part 210-211 – Drugs → Specifications
  - first principle is that ". . . in-process specifications for such characteristics [of in-process material and the drug product] shall be consistent with drug product final specifications . . . ." Accordingly, in-process material should be controlled to assure that the final drug product will meet its quality requirements.
  - The second principle in this regulation further requires that in-process specifications ". . . shall be derived from previous acceptable process average and process variability estimates where possible and determined by the application of suitable statistical procedures where appropriate."

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Regulatory Guidance for Pharma

- Part 210-211 – Drugs → Continued Process Verification
  - Section 211.180(e) requires that information and data about product quality and manufacturing experience be periodically reviewed to determine whether any changes to the established process are warranted.

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New Process Validation Guideline

- A successful validation program depends upon information and knowledge from product and process development. This knowledge and understanding is the basis for establishing an approach to control of the manufacturing process that results in products with the desired quality attributes. Manufacturers should:
  - **Understand the sources** of variation
  - **Detect the presence and degree** of variation
  - **Understand the impact** of variation on the process and ultimately on product attributes
  - **Control** the variation in a manner commensurate with the risk it represents to the process and product

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New Process Validation Guideline

- Manufacturers should use ongoing programs to collect and analyze product and process data to evaluate the state of control of the process. These programs may identify process or product problems or opportunities for **process improvements** that can be evaluated and implemented…

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Other Guidelines

- General Principles of Software Validation
- GAMP
- ISO
- ICH
- GHTF
- Compendia
- Industry Standards

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Understanding, Controlling and Reducing variation during validation

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Variation Understanding and Control with auxiliary processes as input to validation

- Calibration
  - Measurement System Analysis
    - Gage R&R studies aside from regular calibration activities such as Precision, Accuracy, Bias, Linearity, etc.
    - Comparison against AIAG guidelines to determine decision points and improvement opportunities.
  - Scheduled calibration activities.
  - Without precise measurement, the rest of the efforts towards variation control are futile!!
    - Measurements could be biased by operator or instrument.

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Variation Understanding and Control with auxiliary processes as input to validation

- Project Plans (Validation Plans)
  - the studies or tests to use,
  - the criteria appropriate to assess outcomes,
  - the timing of qualification activities,
  - the responsibilities of relevant departments and the quality unit, and
  - the procedures for documenting and approving the qualification.
- Documented R&D Processes and Results
  - Technology Design Transfer
  - Development Studies and Reports
  - V&V during product development

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Variation Understanding and Control with auxiliary processes as input to validation

- Maintenance
  - Scheduled and preventive
  - Effective
- Documentation and Procedures
  - Document Change Control
  - Good documentation practices
  - Guidelines
  - Mistake proofing
- CAPA
  - Ishikawa
  - Flowcharts
  - Technical Investigations

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Variation Understanding and Control with auxiliary processes as input to validation

- Training
  - Effectiveness measurements of training
- Risk Management:
  - With a lifecycle approach to process validation that employs risk based decision making throughout that lifecycle, the perception of criticality as a continuum rather than a binary state is more useful.
  - Design FMECA
  - Process FMECA

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Understanding variation during Design Phase

- Laboratory or Pilot Scale Models
  - Computer Aided Design
- DOE
  - Reveal relationships, including multivariate interactions, between the variable inputs (e.g., component characteristics or process parameters) and the resulting outputs (e.g., in-process material, intermediates, or the final product).
  - Risk analysis tools can be used to screen potential variables for DOE studies.
  - ANOVA
    - One factor studies, two way.
    - AIAG guidelines can be used as decision points.

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Understanding variation during Design Phase

- Hypothesis Tests
  - Comparison between populations or samples.
  - Variation can be measured and decision points can come from a comparison against the “ideal or required” state.
- Flow Charts / Diagrams
  - Understanding process sequence.
  - Elimination of non-value added processes.
  - Sources of variation can be tracked down to the originating steps for correction and improvement.
- Correlation / Regression
  - Influence of one variable over another.
  - A way to trace the original source of variation and choosing between multiple variables.

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Variation Control during Qualification Phase

- Homogeneity within a batch (or a lot) and consistency between batches (or lots) are goals of process validation activities. Validation offers assurance that a process is reasonably protected against sources of variability that could affect production output, cause supply problems, and negatively affect public health.
- For all three phases, IQ, OQ, and PQ, based on product/process requirements:
  - Determine what to verify/measure
  - Determine how to verify/measure
  - Determine how many to verify/measure, i.e. statistical significance
  - Determine when to verify/measure
  - Define acceptance/rejection criteria
  - Define required documentation

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Variation Control during Qualification Phase

- Sampling Process
  - Representative → Process knowledge → Identify the possible sources of variation, understand process behavior:
    - Rational subgrouping
  - Random → Assure through validation procedure.
    - Objective measures (statistical significance).
    - Eliminate noise and uncontrollable variability.
- Acceptance Criteria
  - A description of the statistical methods to be used in collecting and analyzing data (e.g., statistical metrics defining both intra-batch and inter-batch variability).
  - Capability Indices pre-determined for each process.

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Variation Control during Qualification Phase

- IQ – Verification of the design of the facility and qualification of the equipment and utilities
  - Selecting utilities and equipment construction materials, operating principles, and performance characteristics based on whether they are appropriate for their specific uses.
  - Verifying that utility systems and equipment are built and installed in compliance with the design specifications (e.g., built as designed with proper materials, capacity, and functions, and properly connected and calibrated).

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Variation Control during Qualification Phase

- Possible sources of variability address by IQ:
  - Measurement
  - Method
  - Machine
  - “Mother Nature” (Environment)

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Variation Control during Qualification Phase

- OQ - Verification of the design of the facility and qualification of the equipment and utilities related to their operational limits.
  - Verifying that utility systems and equipment operate in accordance with the process requirements in all anticipated operating ranges. This should include challenging the equipment or system functions while under load comparable to that expected during routine production. It should also include the performance of interventions, stoppage, and start-up as is expected during routine production. Operating ranges should be shown capable of being held as long as would be necessary during routine production.

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Variation Control during Qualification Phase

- Possible sources of variability addressed by OQ:
  - Measurement
  - Machine
  - Man
  - Material
  - Method

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Variation Control during Qualification Phase

- PQ - process performance qualification (PPQ)
  - The PPQ combines the actual facility, utilities, equipment (each now qualified), and the trained personnel with the commercial manufacturing process, control procedures, and components to produce commercial batches. A successful PPQ will confirm the process design and demonstrate that the commercial manufacturing process performs as expected.

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Variation Control during Qualification Phase

- Possible sources of variability address by PQ:
  - Man
  - Method
  - “Mother Nature” (Environment)
  - Material

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Variation Control and Reduction during Continued Verification Phase

- Continued Verification Phase:
  - The goal of the third validation stage is continual assurance that the process remains in a state of control (the validated state) during commercial manufacture. A system or systems for detecting unplanned departures from the process as designed is essential to accomplish this goal. **Adherence to the CGMP requirements,** specifically, **the collection and evaluation of information and data about the performance of the process,** will allow **detection of undesired process variability**. Evaluating the performance of the process identifies problems and determines whether action must be taken to correct, anticipate, and prevent problems so that the process remains in control.

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Variation Control and Reduction during Continued Verification Phase

- Continued Verification Plans (Control Plans) - Deliverable
  - Process controls address variability to assure quality of the product. Controls can consist of **material analysis and equipment monitoring** at significant processing points (§ 211.110(c)). Decisions regarding the type and extent of process controls can be aided by earlier risk assessments, then **enhanced and improved as process experience is gained**. (Iterative approach)
  - The planned commercial production and control records, which contain the operational limits and overall strategy for process control, should be carried forward to the next stage for confirmation.

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Variation Control and Reduction during Continued Verification Phase

- CVRs, Trend Data Sheets - Deliverables
  - Considerations for the duration of the heightened sampling and monitoring period could include, but are not limited to, volume of production, process complexity, level of process understanding, and experience with similar products and processes.
  - Variation can also be detected by the timely assessment of defect complaints, out-of-specification findings, process deviation reports, process yield variations, batch records, incoming raw material records, and adverse event reports.
  - Change Control analysis to determine patterns / trends.

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Variation Control and Reduction during Continued Verification Phase

- Tools
  - Statistical Process Control Charts
    - For either variables or attributes
    - Decision making tools (trending)
    - Process should look good but not too good to be in control.
  - Pre-Control
    - Easy to implement.
    - Easy to execute.
  - Process Capability
    - Capability indices
    - PPMs

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A word on PAT

- Process Analytical Technology – Solution offered by FDA for pharma to perform QbD
  - Reducing production cycle times by using on-, in-, and/or at-line measurements and controls
  - Preventing rejects, scrap, and re-processing
  - Real time release
  - Increasing automation to improve operator safety and reduce human errors
  - Improving energy and material use and increasing capacity
  - Facilitating continuous processing to improve efficiency and manage variability

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What to do with Legacy?

- Manufacturers of legacy products can take advantage of the knowledge gained from the original process development and qualification work as well as manufacturing experience to continually improve their processes. → Start in Phase 3: Continued Process Verification and go backwards.
- Risk based approach
- Remediation plans
  - Can't do it immediately, but over time in a planned manner.
- Continuous Improvement

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QA
Experience Sharing

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Thank You

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